

ImageMagick, ExifTool & LibreOffice

Setup guide for the Alfresco Transform Core "All-In-One" (AIO) engine
on a local, non-Docker (zip-based) Windows install of Alfresco Content Services

Covers: why these tools are needed, how to install them, and how to point the transform engine at them.

1. Why these tools are needed

The Alfresco Transform Core AIO engine (`alfresco-transform-core-aio-*.jar`) generates thumbnails, previews, and extracted metadata for uploaded content. Three of its transformers shell out to **external native binaries** rather than doing the work in pure Java:

Tool	Used by	Purpose
ImageMagick	<code>ImageMagickContentTransformer</code>	Resizing / format-converting images for thumbnails and previews
ExifTool	Tika-based metadata transformer (<code>AbstractTikaTransformer</code>)	Extracting EXIF / image metadata for the repository's metadata extraction pipeline
LibreOffice	<code>LibreOfficeTransformer</code> (via JODConverter)	Converting Office documents (docx, xlsx, pptx, odt, ...) to PDF/other formats for previews

Docker vs. bare JAR: The official Alfresco Docker image for this T-Engine bundles these binaries inside the container filesystem (installed via the image's Dockerfile). When you instead run the engine as a bare `java -jar` process on Windows, none of that comes with it — the JAR only contains Java code. Without the binaries installed and referenced correctly, transforms fail with errors such as: `Cannot run program "/usr/bin/convert": CreateProcess error=2` (a hardcoded Linux path that doesn't exist on Windows).

2. Installing the tools (Windows, via `winget`)

2.1 ImageMagick

```
winget install ImageMagick.ImageMagick --silent --force
```

The first attempt without `--force` can fail with `"You cancelled the installation... exit code: 2"` even with `--silent`, due to an unhandled elevation/UAC prompt. Re-running with `--force` resolves it.

ImageMagick 7 installs as `magick.exe` (the legacy `convert.exe` binary is gone), but it retains full compatibility with classic `convert`-style positional arguments, e.g.:

```
magick.exe input.jpg -resize 100x100 output.jpg
```

2.2 ExifTool

The package ID `PhilHarvey.ExifTool` does not exist in the winget registry (`"No package found matching input criteria"`). Find the correct ID first:

```
winget search exiftool
```

The correct package is published as `OliverBetz.ExifTool`:

```
winget install OliverBetz.ExifTool
```

2.3 LibreOffice

```
winget install TheDocumentFoundation.LibreOffice
```

LibreOffice is required because the T-Engine's `LibreOfficeTransformer` drives it headlessly through JODConverter to convert Office documents — the tool needs LibreOffice's `program\soffice.exe` to exist under a normal install root.

3. Locating the installed binaries

Winget install locations can vary (system-wide under `Program Files`, or per-user under `AppData\Local\Programs`). Confirm the paths after installing, e.g.:

```
winget list --id OliverBetz.ExifTool

find "C:/" -iname "exiftool.exe" 2>/dev/null

find "C:/Program Files" -iname "magick.exe" 2>/dev/null
```

Typical results on Windows 11:

Tool	Example install path
ImageMagick	<code>C:\Program Files\ImageMagick-7.1.2-Q16-HDRI\magick.exe</code>
ExifTool	<code>C:\Users\<you>\AppData\Local\Programs\ExifTool\ExifTool.exe</code>
LibreOffice	<code>C:\Program Files\LibreOffice\program\soffice.exe</code>

4. Verifying each install works standalone

```
"C:\Program Files\ImageMagick-7.1.2-Q16-HDRI\magick.exe" input.jpg -resize 100x100 resized.jpg
```

```
"C:\Users\<you>\AppData\Local\Programs\ExifTool\ExifTool.exe" -ver
```

```
"C:\Program Files\LibreOffice\program\soffice.exe" --version
```

5. Configuring the Transform Engine to use them

The T-Engine's `application-default.yaml` exposes each tool's location as an overridable Spring property (`${ENV_VAR:default}` syntax), read from an environment variable at JVM startup:

Spring property	Environment variable	Notes
<code>transform.core.imagemagick.exe</code>	<code>IMAGEMAGICK_EXE</code>	Full path to <code>magick.exe</code>
<code>transform.core.imagemagick.root</code>	<code>IMAGEMAGICK_ROOT</code>	ImageMagick install root
<code>transform.core.imagemagick.dyn</code>	<code>IMAGEMAGICK_DYN</code>	ImageMagick shared-library folder
<code>transform.core.libreoffice.path</code>	<code>LIBREOFFICE_HOME</code>	LibreOffice install root (the folder containing <code>program\soffice.exe</code> , not the exe itself)

ExifTool is the exception. Its Windows command (`transform.core.tika.exifTool.windowsOS` in

`AbstractTikaTransformer`) is a **hardcoded literal string** in the JAR, not an `${ENV_VAR:default}` placeholder

— it cannot be overridden with an environment variable. The only way to satisfy it is to make the plain command

`exiftool` resolvable, i.e. add ExifTool's install folder to the system/process `PATH`.

6. Relaunching the T-Engine with everything wired up

```
set IMAGEMAGICK_EXE=C:\Program Files\ImageMagick-7.1.2-Q16-HDRI\magick.exe

set IMAGEMAGICK_ROOT=C:\Program Files\ImageMagick-7.1.2-Q16-HDRI

set IMAGEMAGICK_DYN=C:\Program Files\ImageMagick-7.1.2-Q16-HDRI

set LIBREOFFICE_HOME=C:\Program Files\LibreOffice

set PATH=%PATH%;C:\Users\<you>\AppData\Local\Programs\ExifTool

java -jar "C:\
\AlfrescoWork\alfresco_install_location\supporting_jar_files\alfresco-
transform-core-aio-5.4.5-A.1.jar" --server.port=8090
```

Only the T-Engine JVM process needs restarting to pick these up. Tomcat/Alfresco (a separate JVM) talks to the T-Engine purely over stateless HTTP calls to `http://localhost:8090/transform` for every request — there's no persistent connection or cached binary state, so it does **not** need to be restarted.

7. Verifying end-to-end

1. Confirm the T-Engine starts cleanly with no ImageMagick/ExifTool/LibreOffice errors in its console output.
2. Upload a new image into Alfresco Share and open its detail page — a thumbnail should render without the previous `imagemagick returned a 400 status` error.
3. Upload a new Office document (e.g. .docx) and confirm a PDF-style preview renders.

Documents/images uploaded *before* this fix will already be marked as failed renditions and generally won't retry automatically — re-upload them, or trigger the rendition again, to see the fix apply.